Hongyue Yu

Email: Natt1e@buaa.edu.cn | Personal Homepage | Github

EDUCATION

Beihang University (BUAA) Sep 2024 – Jul 2027 (*expected*)
Master of Engineering National College for Excellent Engineers

- Advisor: *Prof. Yuan Yuan*

Beihang University (BUAA) Sep 2020 - Jul 2024
Bachelor of Computer Science and Engineering School of Computer Science and Engineering

- GPA: 3.75/4.0
- First Class Scholarship of Beihang University(2021)
- Competition Scholarship of Beihang University(2022)

RESEARCH INTERESTS

Automatic software engineering; AI for software engineering.

SELECTED PUBLICATIONS(Full Publications in Personal Homepage)

1. Kefan Li, Yuan Yuan, **Hongyue Yu**, et al. " CoCoEvo: Co-Evolution of Programs and Test Cases to Enhance Code Generation," *IEEE Transactions on Evolutionary Computation*(**TEVC**), May 2025. [arxiv]

✓ Proposed an LLM-based co-evolutionary framework that jointly generates and iteratively refines programs and test cases without relying on predefined test suites. CoCoEvo combines LLM-driven crossover and mutation, dynamic crossover-rate scheduling, coverage-guided test generation, and Pareto-based multi-objective test selection to improve both code correctness and test quality.

2. Kefan. Li**, Hongyue Yu**, and Yuan Yuan. " Escaping the Self-Repair Trap: Improving Test Oracle Generation via Dual-Context Awareness," in *IEEE/ACM International Conference on Automated Software Enginerring* (**ASE 2026**), Mar 2026. [arxiv]

✓ Identified the Self-Repair Trap, a failure mode in LLM-based test oracle generation where iterative execution-driven repair increasingly favors assertions that are easy to satisfy but less effective at revealing faults. Proposed DCAware, a computationally efficient, non-iterative framework that combines structured static program context with selectively retrieved dynamic execution states to improve oracle grounding.

3. **Hongyue Yu,** Kefan. Li, et al. "TDD-Agent: Test-Driven Reasoning for Code Generation," *Conference on Empirical Methods in Natural Language Processing* (**EMNLP 2026**), May 2026. [arxiv]

✓ Proposed a single-agent framework that operationalizes test-driven development for LLM-based code generation by generating tests before implementation and iteratively co-refining both tests and code using execution feedback. TDD-Agent treats tests as reasoning artifacts rather than fixed post-hoc validators, enabling models to identify implementation defects and correct flawed test assumptions throughout the process. Evaluations on both function-level and the repository-level benchmarks showed consistent improvements over reasoning, retrieval, and agent-based baselines.

4. Kefan Li**, Hongyue Yu**, et al., "Passing is Not Roof: How LLM Agents Decide from Self-Generated Tests", submitted to *International Conference on Software Engineering* (**ICSE 2027**). [paper]

5. Kefan Li**, Hongyue Yu**, Yuan Yuan, et al. "Beyond Fixed Tests: Coevolution of Code and Behavioral Constraints", submitted to *IEEE Transactions on Software Engineering*(**TSE**). [paper]

INTERNSHIPS

Tokfinity AI, Beijing, China Dec 2025-Mar 2026

- Contributed to InfCode, which achieved 1st on Multi-SWE-bench(Java/C++) leaderboards.

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4Paradigm, Beijing, China Apr 2025-Jul 2025

- Trained an attention-based classification head on CLIP for video moderation, achieving 4-way concurrency with sub-4s latency in a resource-constrained environment, with a 12.7% false-negative rate and 17% false-positive rate.

SKILLS

Languages: English (proficient), Mandarin Chinese (native)

Skills: Python, Java, C/C++, C#, etc

Tech Stack: Pytorch, Vue, MySQL, Git, Docker

Enviroment: Linux, macOS, Windows